

WIL Project Briefing

Python PDF Parsing

Graduate Careers Australia (GCA) is a not-for-profit graduate research organisation that works with universities, other research institutions in Australia and the UK, and industry. GCA is a virtual volunteer organisation affiliated with Australia's national peak body for graduate career advisors.

GCA also works with universities across Australia to supervise student WIL projects. GCA values WIL program roles in career readiness development of graduates and actively collaborates with universities to identify simulated industry projects for these programs. Projects are supervised by GCA, either remotely and/or, preferably, on campus.

PROJECT OVERVIEW

This project focuses on developing a PDF document parsing solution to support other downstream applications. The scope of work is limited to researching, designing, experimenting, and developing Python-based algorithms. These algorithms must be sufficiently generalised to process documents that vary significantly in formatting and document types. The outputs generated by these algorithms will contribute to downstream processing. The extracted data will include metadata, contextual information, and key structural elements of the source documents that support semantic interpretation of the PDF data.

A graphical user interface (UI/UX) is not required for this stage of development. The algorithms will serve as callable Pythonic functions within a client development environment. Environments such as VS Code and extensions may be utilised for development. The final deliverable will consist of well-structured and well-documented Python functions.

This project is focused on knowledge creation and transfer rather than application development. Students will research, learn, experiment, develop and document reusable code (“library functions”) for PDF parsing.

WIL PROJECT

The objectives of this WIL project have been deliberately scoped to ensure meaningful, technically robust outcomes within the constraints of an academic semester. Students, within general approval guidelines and advice, will scope their project and define their own goals. In addition to meeting technical requirements, the assessment will strongly emphasise maintainability and documentation quality, enabling future student teams to extend and refine this foundational work.

NOTES

- For production ... Python only (no Java, LLMs, databases, ...); PDF tool = PyMuPDF
- Text processing tools to be fast, free, license-free, and open source
- A sample test harness and test files will be provided; the test harness (**test_harness.py**) will call functions that extract and normalise data blocks ready for downstream processing:

```
raw_data = extract_raw_data(pdf_path)
logical_doc(s) = split_logical_docs(raw_data)
normalized_doc(s) = normalize(logical_doc)
```

- The task is to process generic documents of varying formats, providing extraction and structural tagging of PDF data
- In addition to the Python code and technical documentation, a brief report is to be prepared for GCA outlining the approach, challenges, and areas for potential development and improvements as an aid to future student teams

TIMETABLE

Week	Activity	Notes/Deliverables
1	GU – WIL Project Introduction	Course introduction and academic requirements ¹
2	Initial Project Meeting Requirements Definition Meeting	Personnel introductions, project overview and project management structure; project specifications ²
3	Project team working	Requirements Definition ²
4		
5		
6	Mid-Project Meeting	Project progress meeting
7	Project team working	
8		
9		
10		
11		
12	Final Presentation Deliverables Handover	Code; Documentation

¹ Students should prepare and submit at end-week 1, a mini-CV (dot point, half-page max.), briefly outlining their IT development skills and experience.

² This task will require the team to conduct research on PyMuPDF and related PDF text and table parsing techniques and to meet with GCA as required. Project meetings may be held remotely or, ideally, face-to-face on campus.

GCA CONTACT

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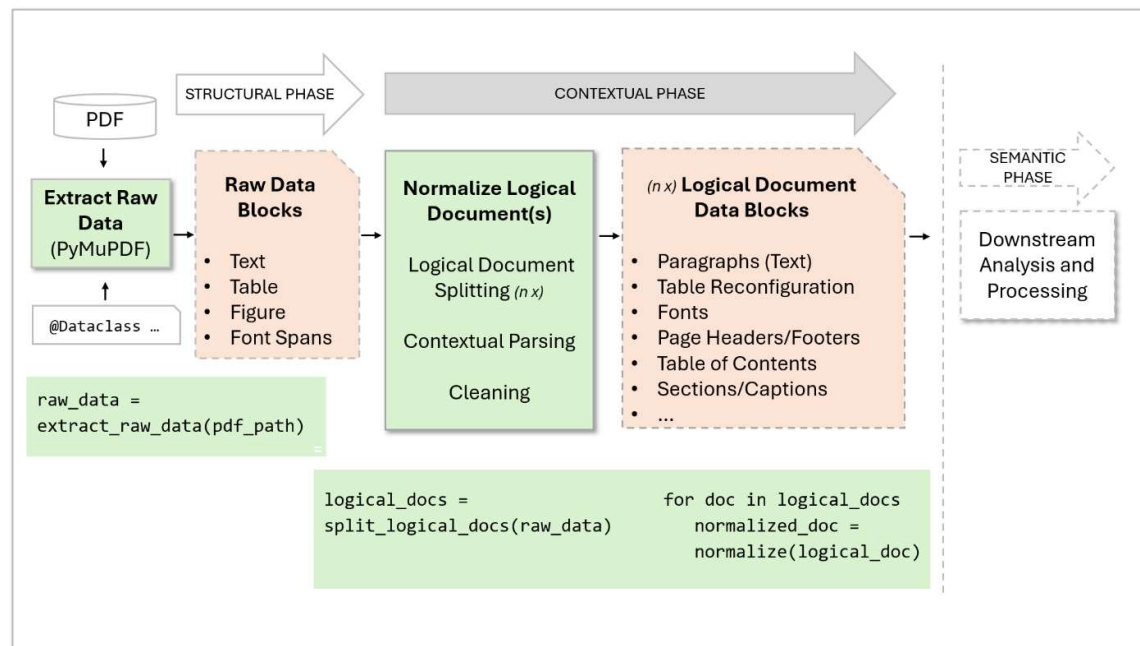
PDF DOCUMENT NOTES

PDF data can be considered to be: Structural, Contextual or Semantic.

Structural characteristics refer to how the document is physically organised—how pages are divided, tables are configured, or headings and paragraphs are separated. For example, a PDF might have a clear table grid, multiple columns, or distinct sections marked by bold headings. Properties of PDF data concern the specific attributes of its elements – bounding box coordinates and font size/style used for text and headings. These properties define the content's visual appearance and technical details.

Contextual data defines how elements relate to one another within the document. The meaning of a table, for instance, can be clarified by its caption or by the section title that precedes/follows it. Text, structured with multi-level hierarchical section headings, can provide context for paragraphs, tables, figures, and other content elements.

Semantic characteristics focus on the content's actual meaning and purpose. This is about understanding what the data represents—such as recognising a table as a structured summary of data, identifying a paragraph as a conclusion, or interpreting a section as a list of options. Semantic understanding enables the extraction of the true value and intent behind the information presented.

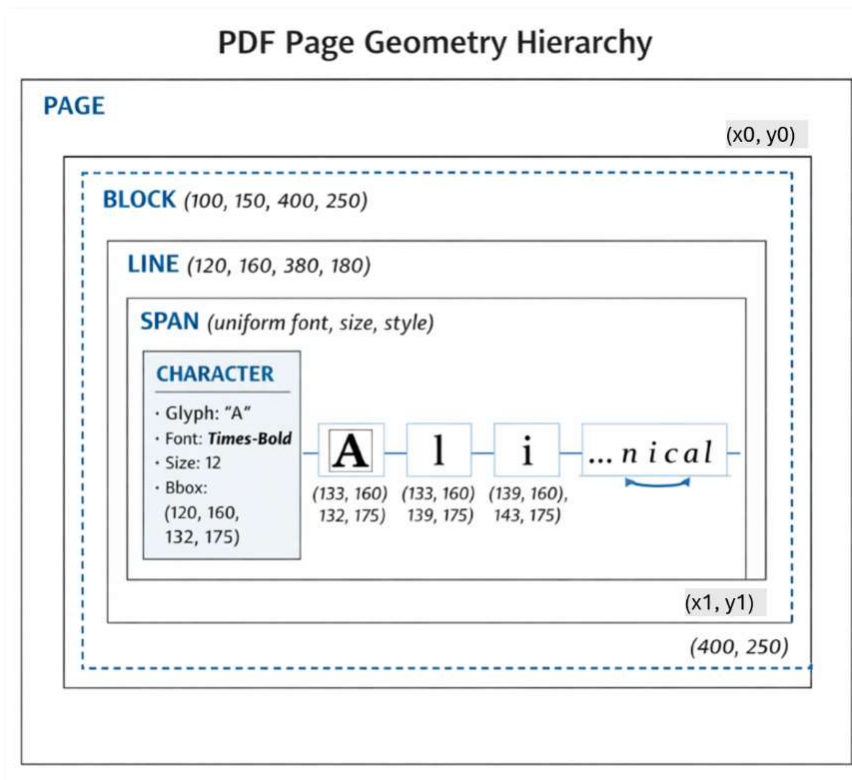


Broadly, this project is positioned in the CONTEXTUAL PHASE with some possibility of modifying the STRUCTURAL PHASE (notably @dataclass definitions) if justified.

PDF FILES

The project will require students to know (and probably learn) something about the internal architecture and geometry of PDF files.

Internally, the content is organised into pages, with each page referencing its associated resources, such as text, fonts and figures.



The content is composed of data blocks, which are the basic units of information—these can represent text, vector graphics, images, or other elements. Each block is associated with coordinates (x0, y0, x1, y1) that define its position and size on the page, allowing precise spatial mapping of content.

Text blocks, for example, contain sequences of characters along with positioning, font, size, and style information, while table or figure blocks may be embedded as streams with their own coordinates and properties.

This internal structure allows PDF extraction tools like PyMuPDF to navigate from the page level down to individual text spans and characters, capturing both the textual content and its exact placement within the document for tasks like layout reconstruction, table extraction, or page header identification. This structure is central to the PDF concept as a portable display format able to be rendered on multiple platforms.

PyMuPDF

PyMuPDF is a Python library that provides fast access to PDF content. Its speed and precision make it ideal for automated document analysis.

PyMuPDF interacts with the internal structure of a PDF, navigating from pages to data blocks and down to individual characters.

Each page object in PyMuPDF provides access to the content streams of that page, which are internally composed of blocks—these can be text, vector graphics, images, or other elements.

Text blocks are further divided into font spans, where each span represents a contiguous sequence of characters that share the same font properties, such as font name, size, and style. These spans also carry coordinate information, allowing precise placement of text. This span-level structure enables accurate reconstruction of document layout, including tables, headers, and footers. Similarly, image and shape blocks retain their spatial positioning, enabling the extraction of figures and diagrams.

